



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech.(CBS) (CBCS) Degree Examinations
 January 2023

22CBS111 – CLOUD SECURITY AND IOT SECURITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is virtualization? Explain with the help of Virtualization Architecture diagram explain different types of Virtualizations.	10	L*2	2	1,2
	b) List and Explain types of Cloud computing based on Deployment model.	10	L2	1	1
2.	a) List and explain different service models of cloud Computing.	10	L2	1	1
	b) What is threat modelling? List and explain different stages of Threat modelling.	10	L2	2	1,2
3.	a) Explain the benefit and limitations of Security Models in IaaS, PaaS, SaaS.	10	L2	1	1
	b) What is CIA in Cyber Security? Explain the steps experiment with ATM, which allows users to access bank balances and other information using CIA triad.	10	L3	2	1,2,3
Unit – II					
4.	a) List different phases of Penetration Testing. Differentiate Vulnerability scan and Penetration testing.	10	L2	3	1
	b) Explain how key management works with a nice illustration.	10	L2	3	1,2
5.	a) List different types of attack. Explain SQL injection attack with the help of neat diagram.	10	L2	3	1,2
	b) Explain the advantages and major challenges of cloud security.	10	L2	3	1
6.	a) Identify and explain Critical concepts for data security in cloud	10	L2	3	1,2
	b) Make use of the diagram and explain the role of Identity Management (IDM) System in online shopping system.	10	L3	3	2
Unit – III					
7.	a) Explain different types of security threats and their attack procedures.	10	L2	4	1
	b) Explain the Security Threats at the Sensing/Perception Layer.	10	L2	4	1,2
8.	a) Outline the Security Requirements in Internet of Things.	10	L3	5	1
	b) List and explain the most important security concerns in IoT four-layer architecture.	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
